

PrivML: Test Outputs

Timestamp: 2025-09-19 20:06:17
Model ID: Model 46

Dataset Path: data/mnist/
Number of Images in Dataset: 10000
Training Size: 75%
Validation Size: 15%
Testing Size: 9%
Encrypted Testing Subset Size: 1%
Data Augmentation: True
Random Seed: 500

Plain Model Training

Epochs: 75
Batch Size: 64
Learning Rate: 0.0001
Optimizer: Adamax
Loss Function: CrossEntropyLoss
Sample Output Count: 6

Plain Model Architecture

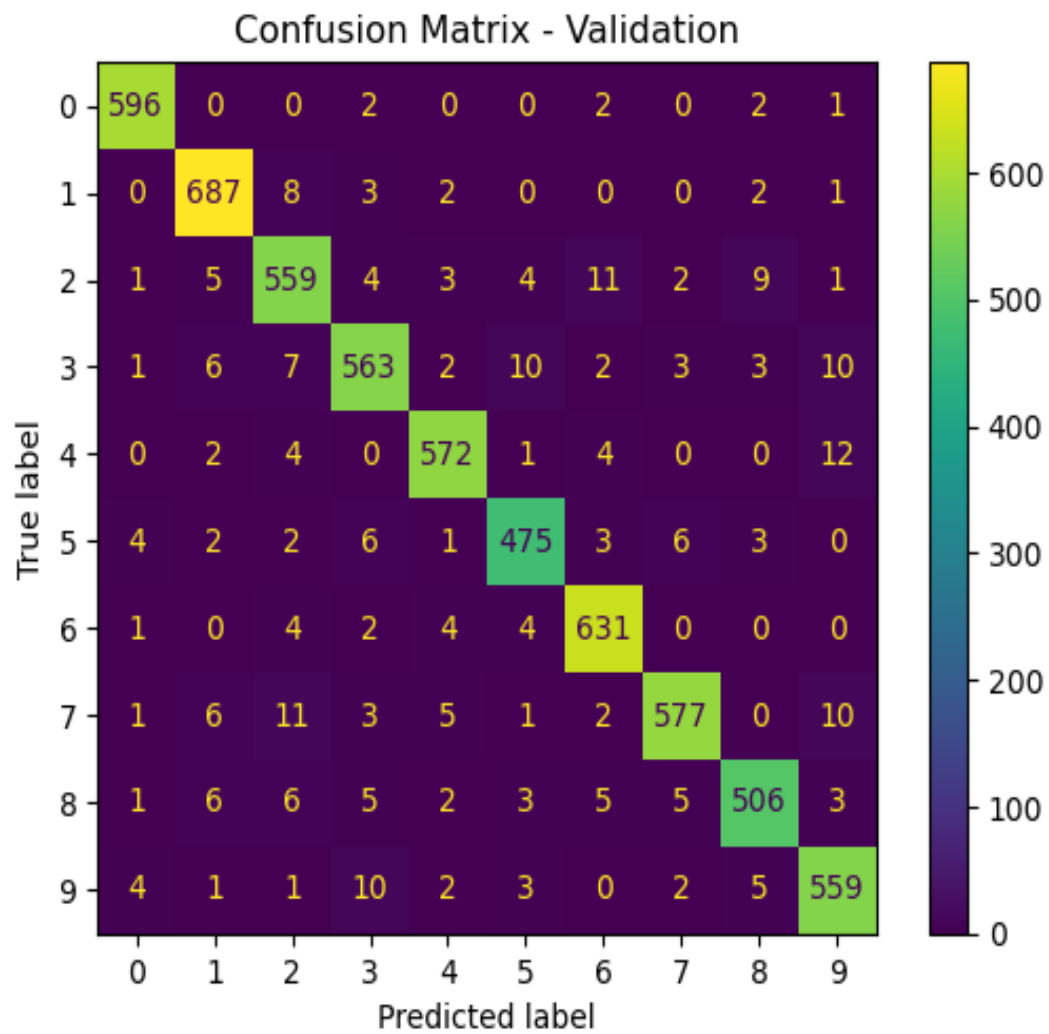
Hidden Layer Size: 256
Output Classes: 10
Conv Kernel Size: 7
Conv Stride: 3
Conv Padding: 0
Dropout Rate: 0.3

FHE Configuration

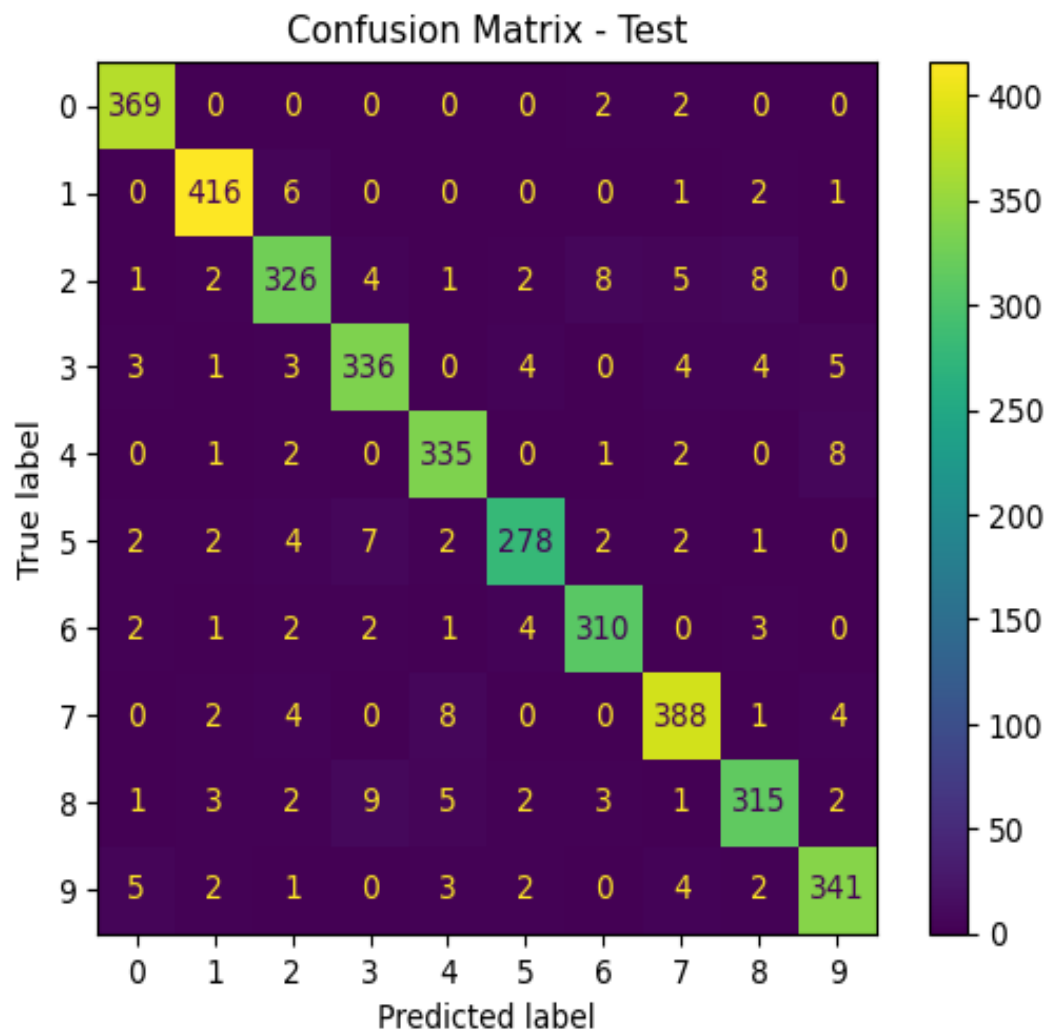
Encryption Scheme: CKKS
Scale (bits): 26
Polynomial Modulus Degree: 8192
Key Generation Level: 31
Decryption Stability Level: 31
Encrypted Loss Function: CrossEntropyLoss
Coefficient Mod Bit Sizes: [31, 26, 26, 26, 26, 26, 26, 31]

Privacy-Preserving ML: ML Metrics Report

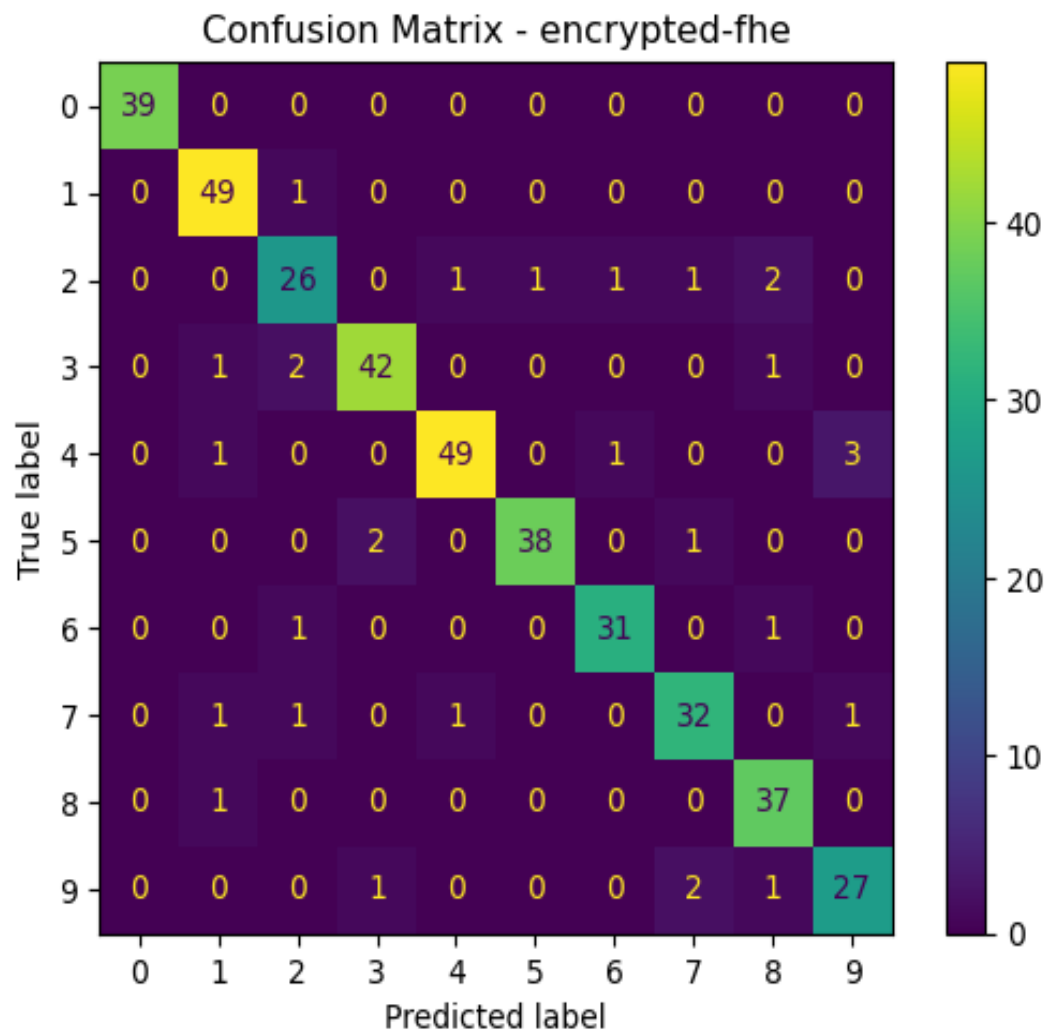
Confusion Matrix - Validation



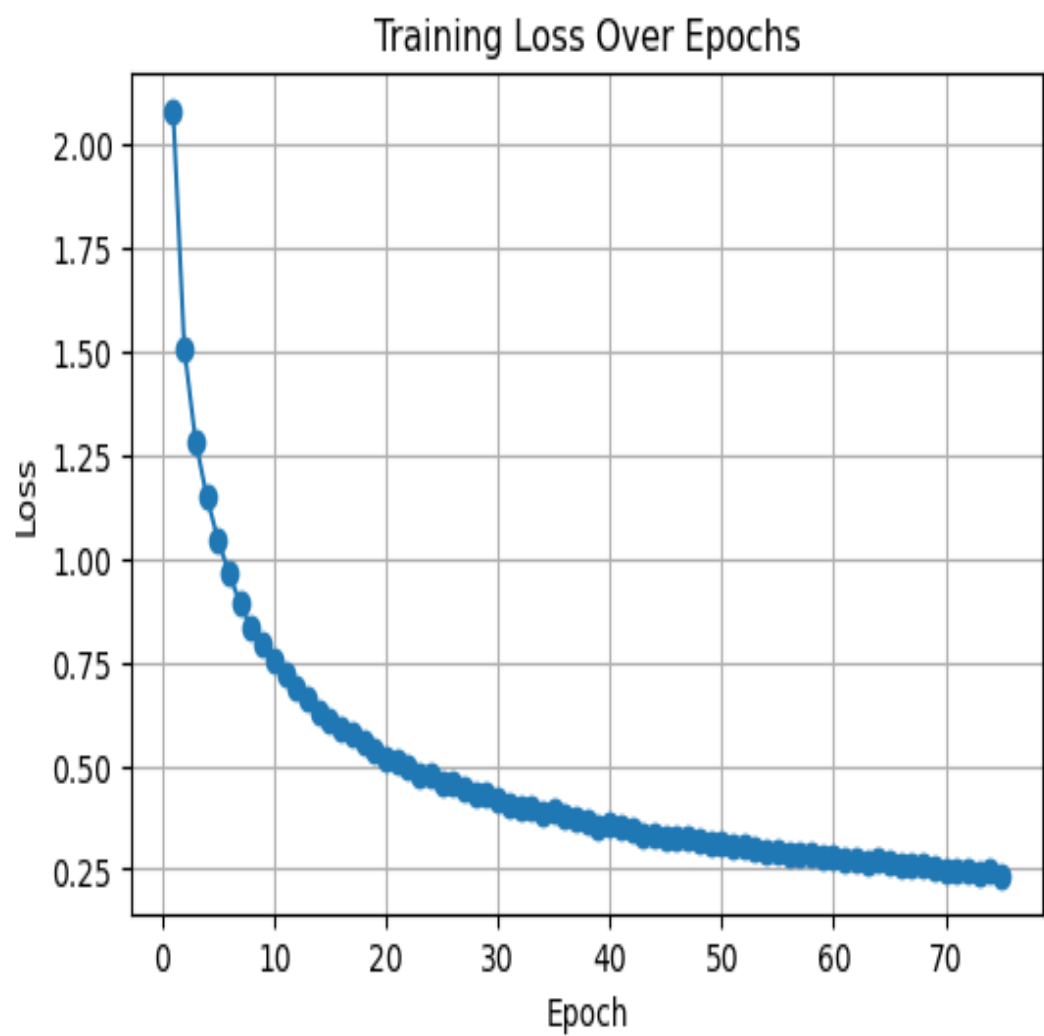
Confusion Matrix - Test



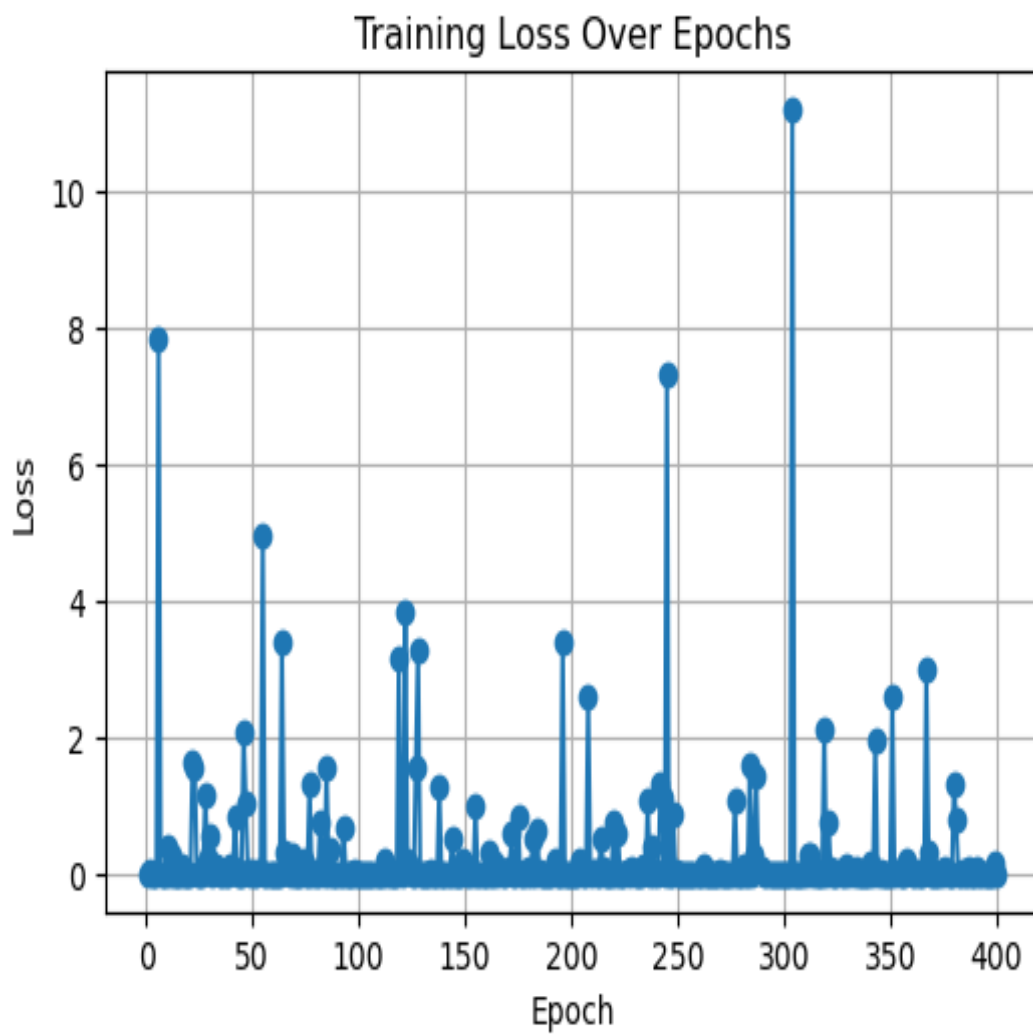
Confusion Matrix - encrypted-fhe



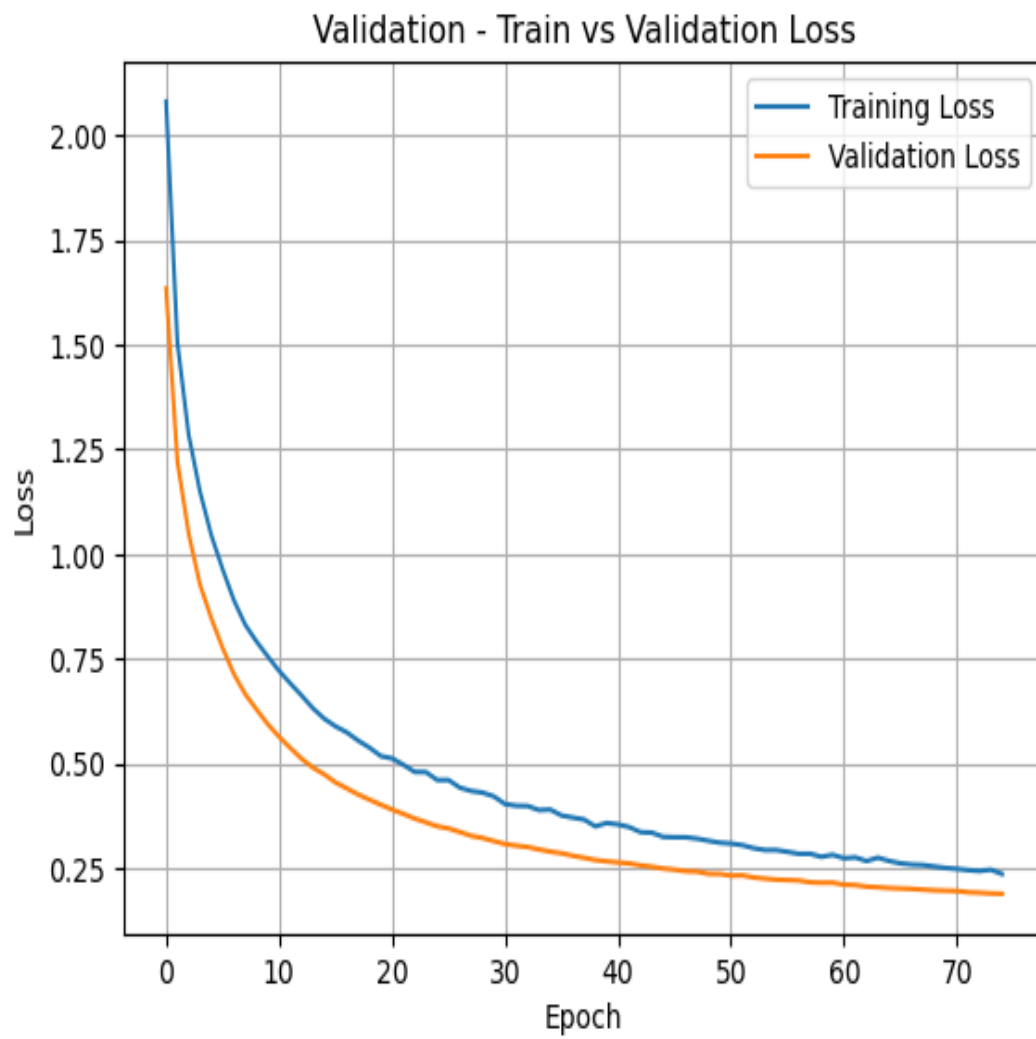
Training Loss - Validation



Training Loss - encrypted-fhe

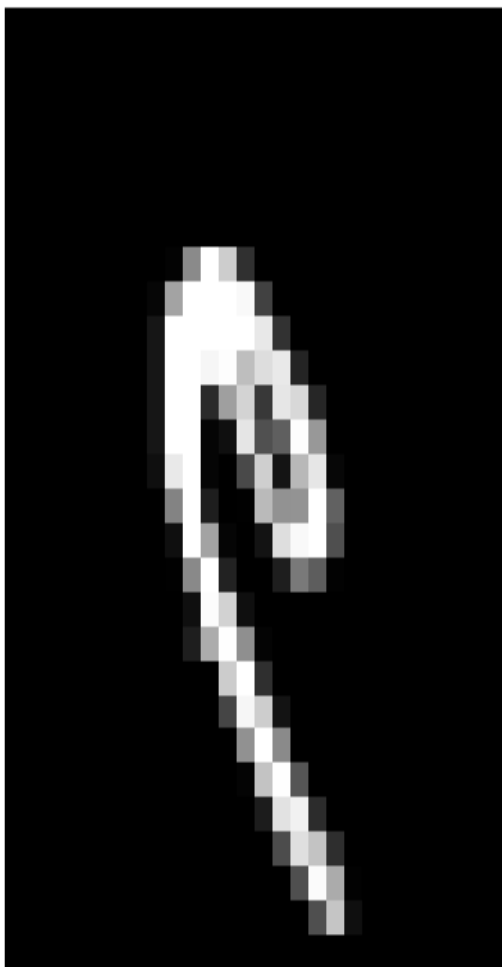


Train vs Validation Loss - Validation

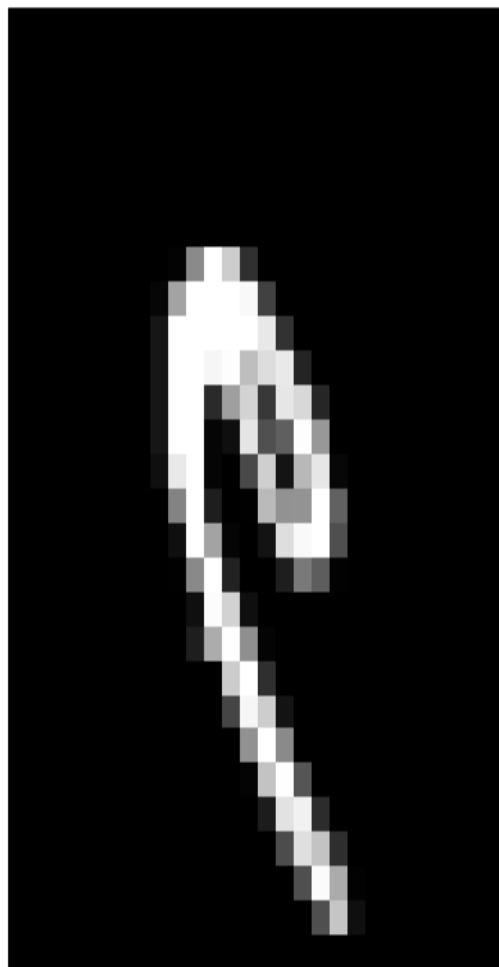


Predicted Images - Predicted Image

Ground Truth: 9

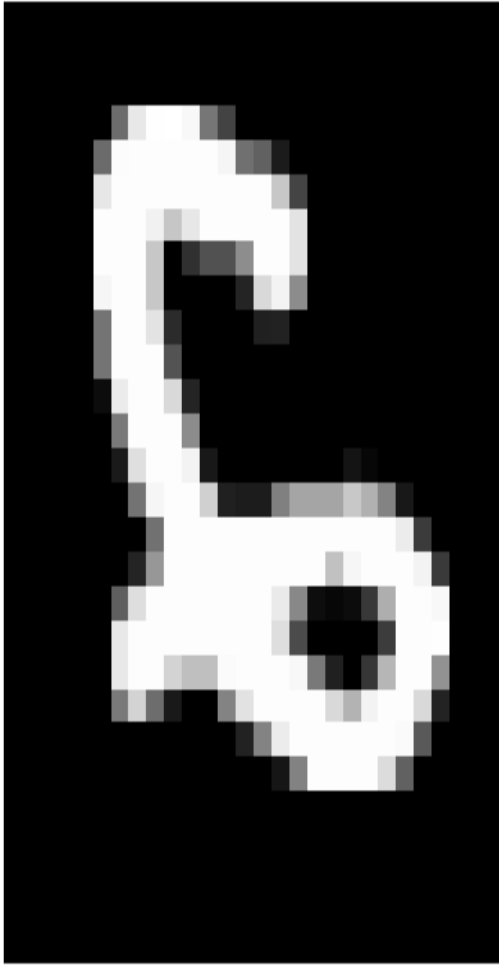


Predicted: 9

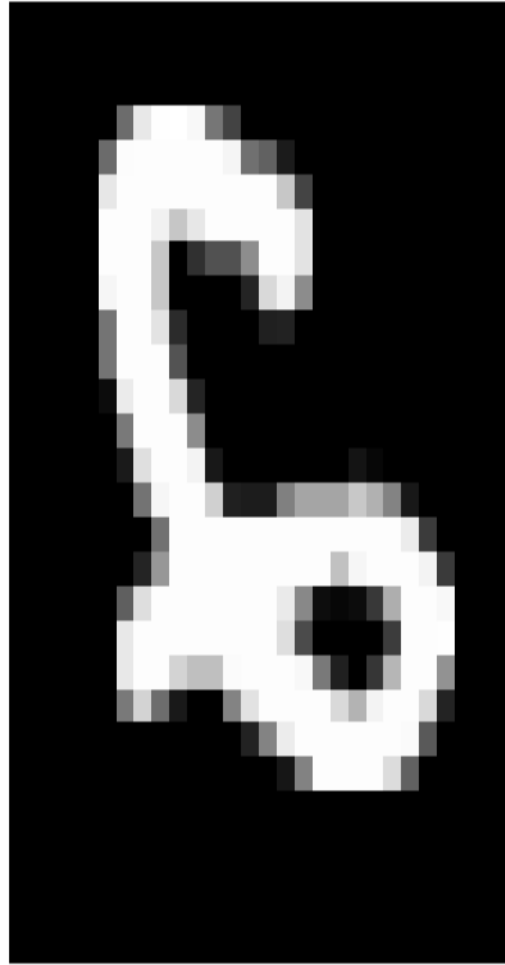


Predicted Images - Predicted Image

Ground Truth: 2

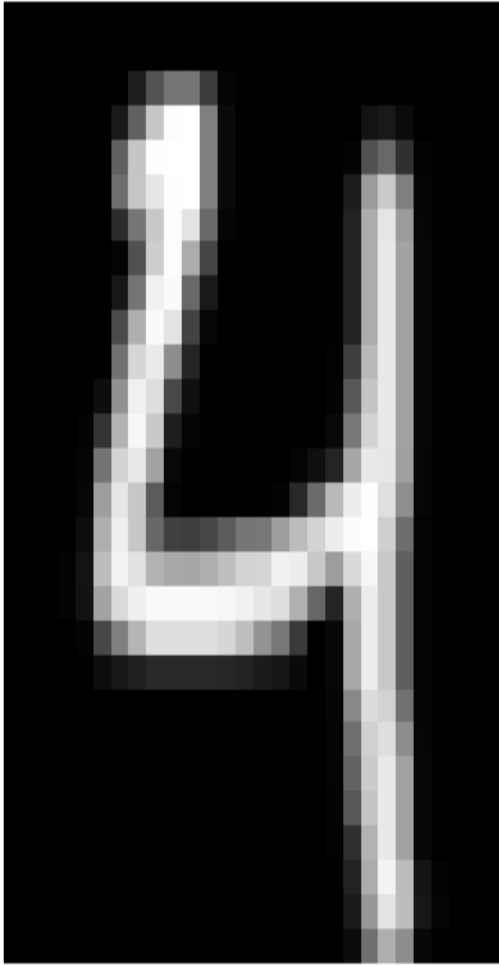


Predicted: 2

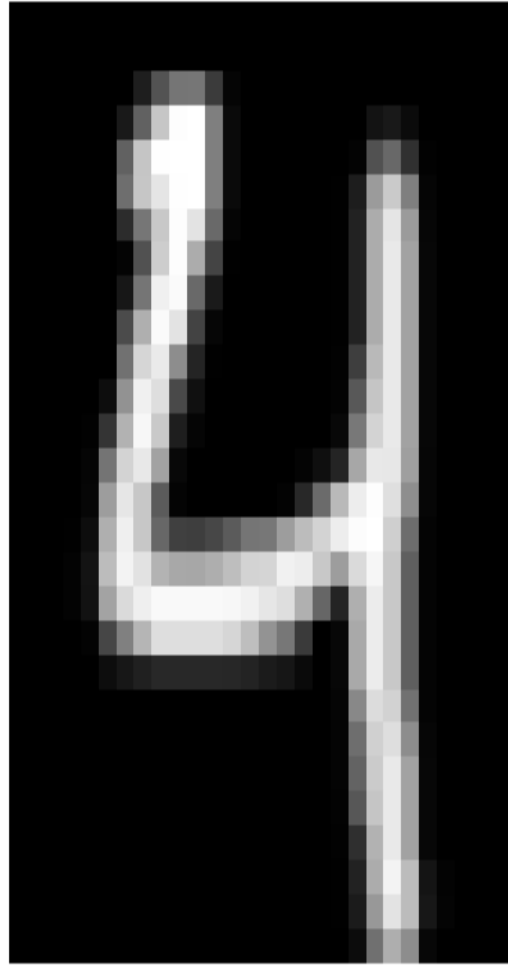


Predicted Images - Predicted Image

Ground Truth: 4

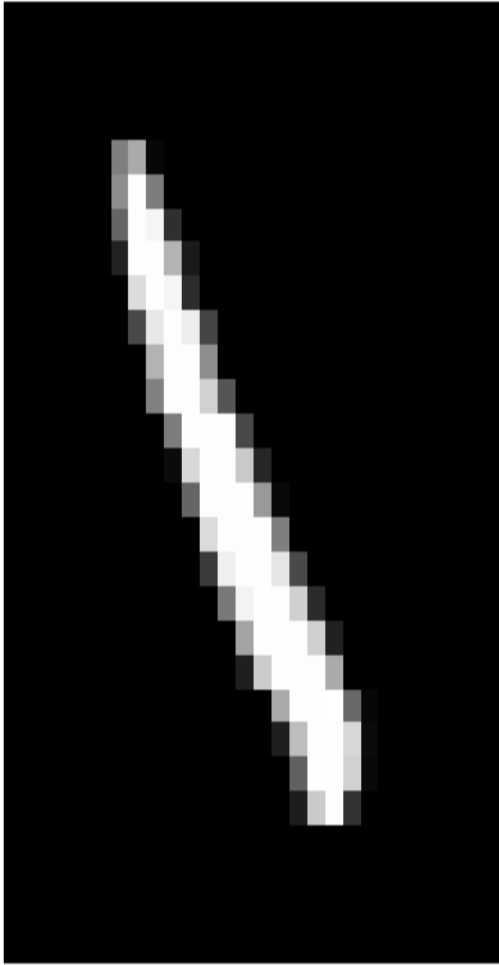


Predicted: 4

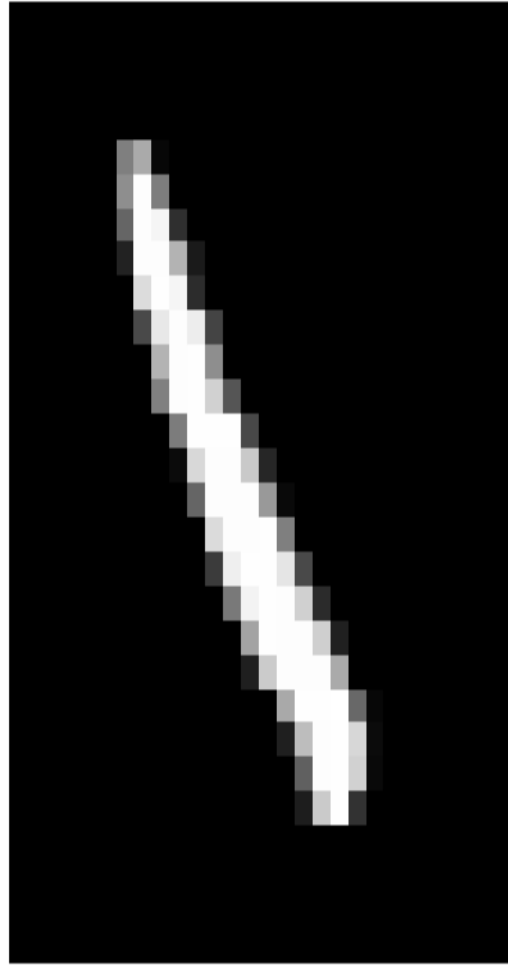


Predicted Images - Predicted Image

Ground Truth: 1

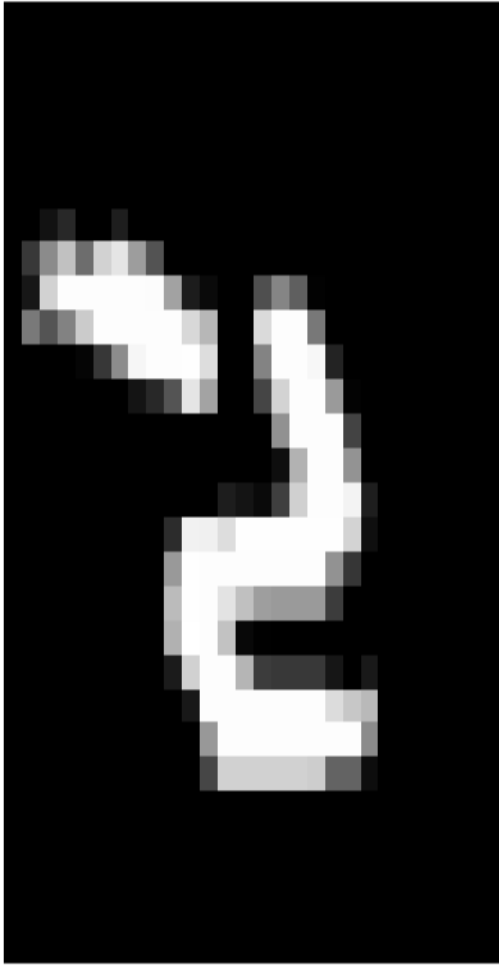


Predicted: 1

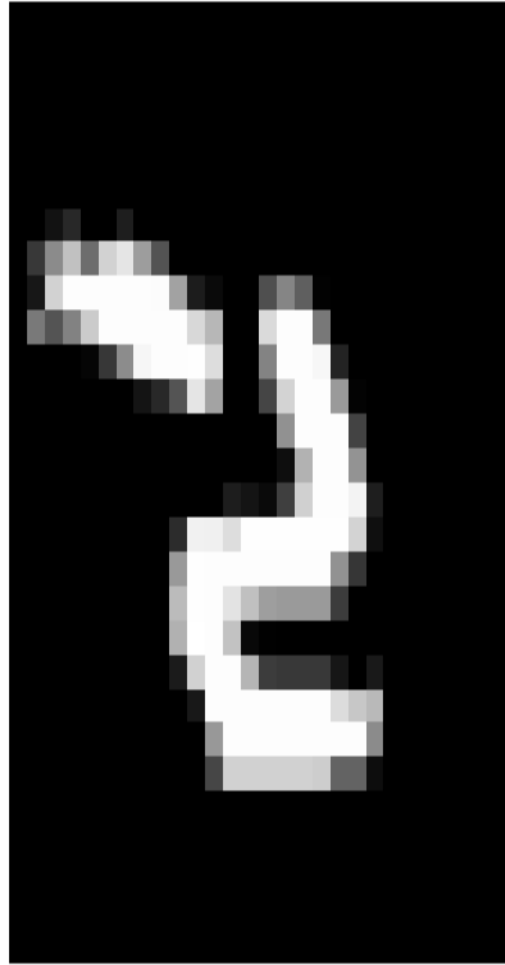


Predicted Images - Predicted Image

Ground Truth: 5

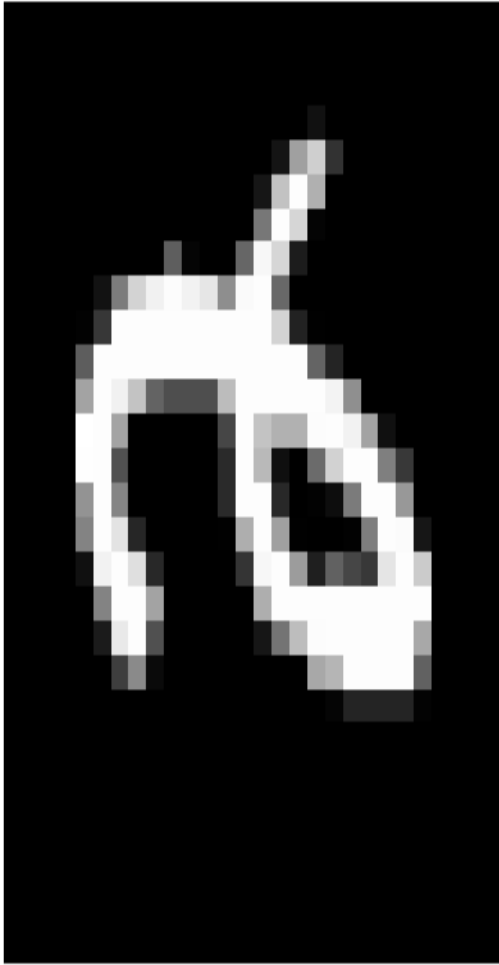


Predicted: 5

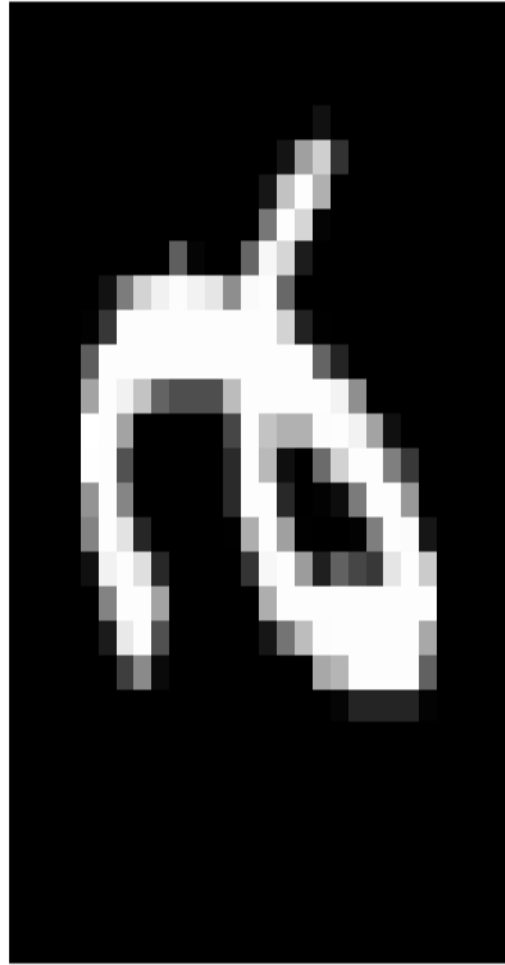


Predicted Images - Predicted Image

Ground Truth: 2



Predicted: 2



Runtime System Resource Usage

encrypted-fhe: OS: macOS-26.0-arm64-arm-64bit

encrypted-fhe: Python Version: 3.10.8

encrypted-fhe: Processor: arm

encrypted-fhe: CPU Usage (%): 4.8

encrypted-fhe: RAM Usage (Used GB): 8.66

encrypted-fhe: RAM Usage (Total GB): 18.0

encrypted-fhe: RAM Usage (%): 62.4

encrypted-fhe: GPU: Not Available

encrypted-fhe: CPU Usage (%): 18.4

encrypted-fhe: RAM Usage (Used GB): 9.26

encrypted-fhe: RAM Usage (Total GB): 18.0

encrypted-fhe: RAM Usage (%): 63.3

encrypted-fhe: GPU: Not Available

encrypted-fhe: CPU Usage (%): 12.8

encrypted-fhe: RAM Usage (Used GB): 8.6

encrypted-fhe: RAM Usage (Total GB): 18.0

encrypted-fhe: RAM Usage (%): 60.8

encrypted-fhe: GPU: Not Available

encrypted-fhe: CPU Usage (%): 15.9
encrypted-fhe: RAM Usage (Used GB): 8.27
encrypted-fhe: RAM Usage (Total GB): 18.0
encrypted-fhe: RAM Usage (%): 61.6
encrypted-fhe: GPU: Not Available
encrypted-fhe: CPU Usage (%): 12.3
encrypted-fhe: RAM Usage (Used GB): 8.27
encrypted-fhe: RAM Usage (Total GB): 18.0
encrypted-fhe: RAM Usage (%): 61.5
encrypted-fhe: GPU: Not Available
encrypted-fhe: CPU Usage (%): 12.6
encrypted-fhe: RAM Usage (Used GB): 8.57
encrypted-fhe: RAM Usage (Total GB): 18.0
encrypted-fhe: RAM Usage (%): 63.2
encrypted-fhe: GPU: Not Available
encrypted-fhe: CPU Usage (%): 12.5
encrypted-fhe: RAM Usage (Used GB): 8.29
encrypted-fhe: RAM Usage (Total GB): 18.0
encrypted-fhe: RAM Usage (%): 61.7
encrypted-fhe: GPU: Not Available
encrypted-fhe: CPU Usage (%): 12.6
encrypted-fhe: RAM Usage (Used GB): 8.29
encrypted-fhe: RAM Usage (Total GB): 18.0
encrypted-fhe: RAM Usage (%): 61.6
encrypted-fhe: GPU: Not Available
encrypted-fhe: CPU Usage (%): 33.6
encrypted-fhe: RAM Usage (Used GB): 8.37
encrypted-fhe: RAM Usage (Total GB): 18.0
encrypted-fhe: RAM Usage (%): 62.0
encrypted-fhe: GPU: Not Available